Algorithm Runtime · Synthetic Lab Sheet

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This synthetic lab compares input size with measured runtime. The chart is illustrative, not a formal complexity proof. Binary search requires sorted input and performs logarithmically many comparisons in the worst case.

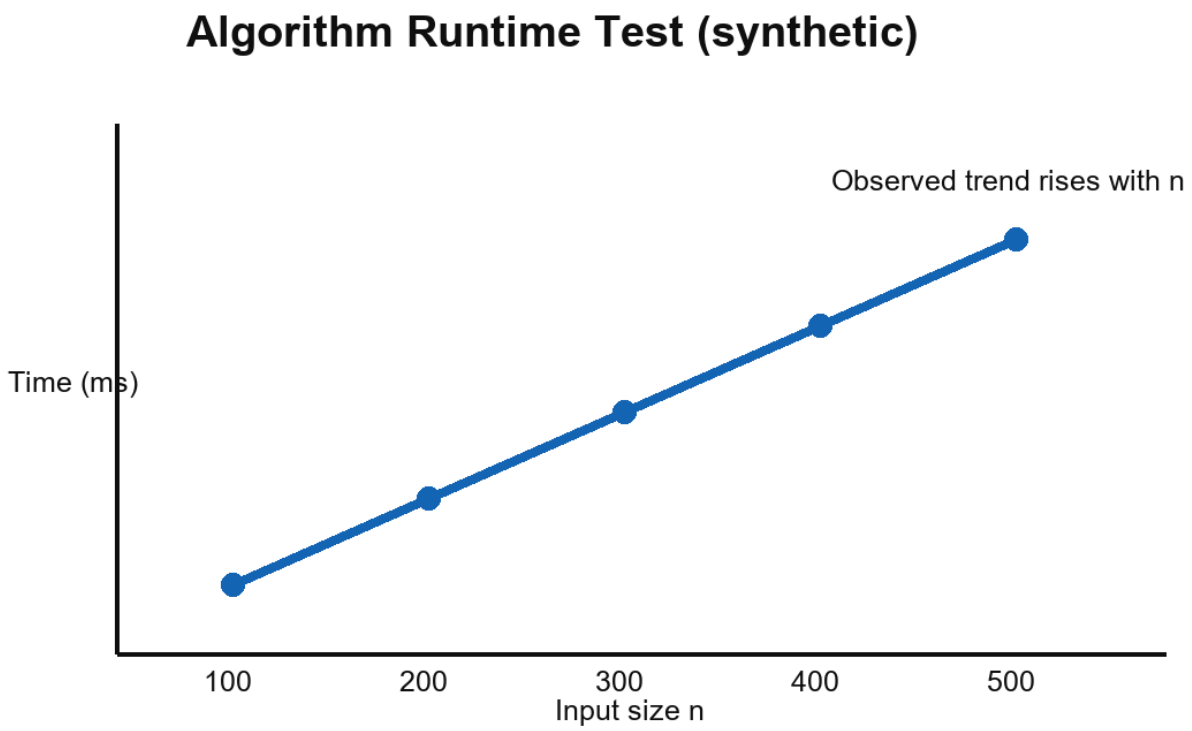


Figure 1: Observed runtime trend across five input sizes.

Interpretation task: identify the horizontal axis, vertical axis, and overall trend. Do not infer an exact asymptotic bound from this chart alone. A controlled benchmark needs repeated trials and documented hardware.